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Department of Mathematics

Calculus II - MA1220

End Term, Date: Jan 03, 2023

Time: 17:30 to 19:30 | Dec - Jan 2023 | Max points: 40

- Attempt all questions.
- Each question carries five points.
- Mobile phones or calculators are not allowed.

1. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Find the value of $f(4)$ using the following identity

Thoram.....?

$$\int_0^{x^2} f(t)dt = x\cos(\pi x).$$

2. Evaluate

$$\int_0^1 x \ln x \; dx.$$

3. Let

$$f(x,y) = \begin{cases} x^2 & x \geq 0, \\ x^3 & x < 0. \end{cases}$$

Find the following limits using the definition (use ε and δ arguments if needed)

(a) $\lim\limits_{(x,y)\to(3,-2)} f(x,y)$

(b) $\lim\limits_{(x,y)\to(0,0)} f(x,y)$

4. Find the maximum and minimum value of $f(x,y) = x^2 + y^2$ on the line $x + y = 1$ using the Lagrange multiplier.

5. Find the local extrema (maxima / minima) of the function $f(x,y) = (x^2 - 4x)\cos(y)$, where $x \in (1,3)$ and $y \in (-\frac{\pi}{4}, \frac{\pi}{4})$.

PTO

6. Let $f(x,y) = x^2 + xy + y^2 + ax + by + c$, find the values of a, b and c giving local minimum at $(2,5)$ and so that $f(2,5) = 11$.

7. Find the surface area of the surface generated by the curve $y = x^2 + 2$ revolving around the x axis from $x = -3$ to $x = 3$.

Integration ??

8. Let $f(x,y) = \dfrac{x\sqrt{|y|}}{\sqrt{x^2+y^2}}$. Determine whether or not the limit $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exists.

Best wishes